

CANDIDATE
NAME

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INDEX NUMBER

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BIOLOGY

9648/03

Applications Paper and Planning Question
Paper 3

20 September 2016
Tuesday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
1	
2	
3	
4	
5	
Total	72

Answer **all** the questions.

- 1
(a) State **three** differences between a genomic DNA library and a cDNA library.

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[3]

Many countries are trying to increase their production of bioethanol in order to fuel a range of vehicles. Most bioethanol is produced from the cellulose cell walls of plants. The cellulose synthase gene have been identified and sequenced.

In initial experiments to increase cellulose production, DNA containing the cellulose synthase gene were inserted into plasmid vectors and subsequently introduced into bacteria cells through transformation. Fig. 1.1 is a diagram of the pUC 19 plasmid map used in the experiments.

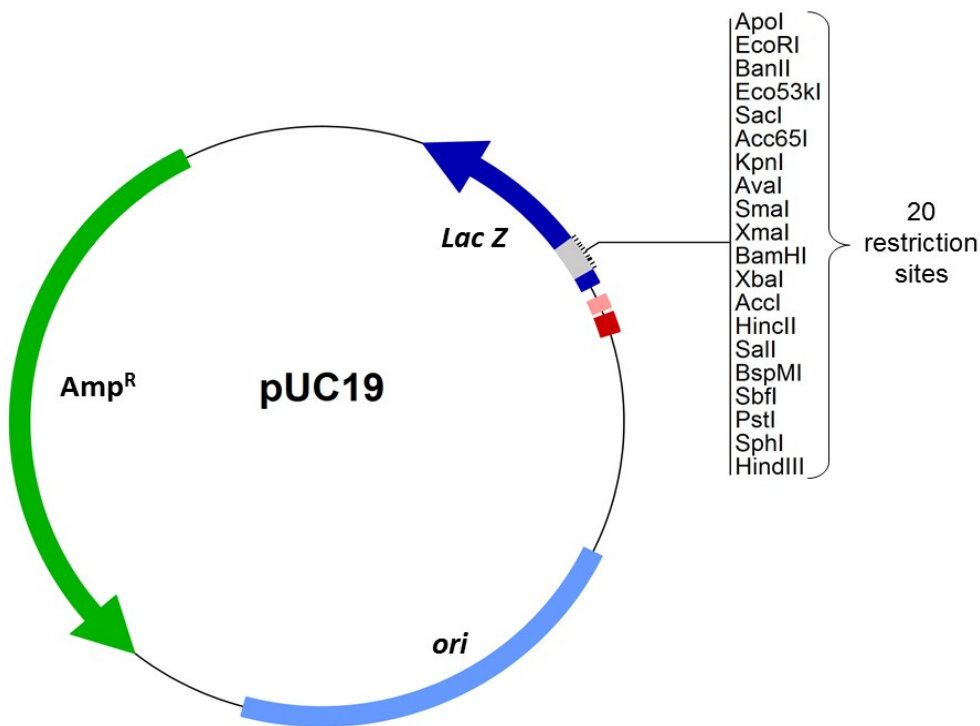


Fig. 1.1

- (b) (i) State and explain **two** visible properties of the pUC 19 plasmid that allow it to be used as a DNA cloning vector.

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[4]

- (ii) Explain how successfully transformed bacterial cells containing recombinant plasmids can be selected for.

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[3]

- (c) The cellulose synthase gene obtained from a plant species has approximately 5000 base pairs (bp). However the length of the gene sequence inserted into bacterial cells was shortened to only approximately 3300bp.

Explain why the length of the gene inserted into bacterial cells was shortened.

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[2]

- (d) Using bacteriophages to introduce the cellulose synthase gene into bacterial cells was thought to improve transformation efficiency. However, a lower than expected proportion of bacterial cells was found to contain the gene after bacterial transduction using bacteriophages.

Explain why using bacteriophages to introduce DNA into bacterial cells was not successful.

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[2]

[Total: 14]

- 2 Antithrombin (AT) is a protein required for the prevention of blood clots in the blood plasma. Patients with hereditary AT deficiency requires AT injection before they undergo any surgery to prevent the formation of blood clot in the veins.

ATryn, a recombinant AT (rAT) is the first medicine produced using genetically engineered animals. rAT is made from the mammary glands of goats that have been genetically modified (GM). The DNA construct introduced into goats comprises of the cDNA of human AT and regulatory elements of the goat beta casein gene, an essential milk protein produced in goat mammary glands. Microinjection was used to insert the DNA construct into a goat zygote which is then transplanted into the uterus of a female goat. GM progeny goat will produce large amounts of AT into its milk which are then isolated and purified.

- (a) Explain why the DNA construct used is a recombinant DNA.

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 [1]

- (b) Suggest why the DNA construct used must include regulatory elements of the goat beta casein gene.

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 [2]

The effectiveness of rAT can be measured in terms of percentage activity compared with normal human blood plasma AT (hpAT). rAT was injected into four patients with AT deficiency (AT activity $\leq 60\%$) and their AT activities were measured every two days after the injection. Fig. 2.1 shows the results of the study.

Table 2.1 shows a comparison between the structure of rAT and hpAT.

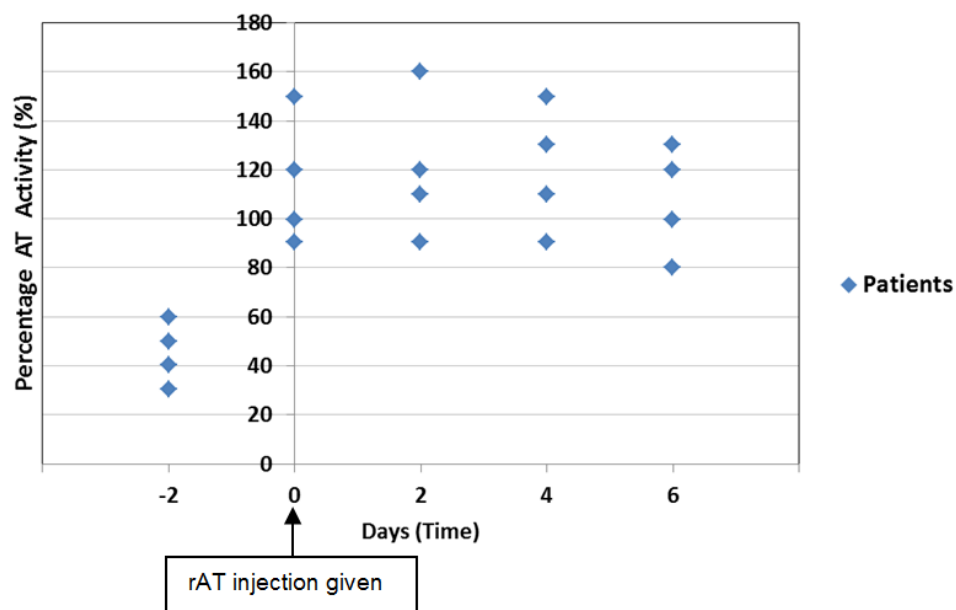


Fig. 2.1

	rAT	hpAT
Primary structure	Single chain of 432 amino acids	Single chain of 432 amino acids
Glycosylation	Glycosylation at 3 asparagine amino acid residues (Asn 96, 155, 192)	Glycosylation at 4 asparagine amino acid residues (Asn 96, 135, 155, 192)
Analysis of carbohydrate chains on the protein	Mainly N-acetyl neuraminic acid	Mainly N-glycolyl neuraminic acid

Table 2.1

(c) Using the information above,

(i) describe the effect of injecting rAT on the patients .

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 [1]

(ii) describe a difference between the structures of rAT and hpAT and the effect of the difference on the patients.

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 [2]

(iii) suggest what resulted in the differences in the structures of rAT and hpAT.

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 [1]

Polymerase chain reaction (PCR) and gel electrophoresis can be used to distinguish between a DNA construct containing both cDNA of human AT and regulatory elements and one which contains only the regulatory elements.

Fig. 2.2 shows the DNA construct used in forming rAT.



Fig. 2.2

- (d) (i) On Fig. 2.2, indicate with **5' → 3'** where the primers will bind in the PCR reaction.

5' and **3'** represents the 5' and 3' ends of the primers respectively.

[1]

- (ii) Explain how gel electrophoresis can be used to distinguish between a DNA construct with both cDNA of human AT and regulatory elements and one which contains only regulatory elements.

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[2]

- (e) A new project, GenomeAsia 100K, aims to sequence the genomes of 100,000 Asians across Asia to tailor-made drugs for Asians, an approach known as precision medicine.

- (i) State **one** benefit of precision medicine.

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[1]

- (ii) While there are many benefits of precision medicine, some people are apprehensive that the project will unravel genetic differences in ethnicity “making some people a lesser man and others a better man”

Explain why the project may result in the ethical issue of “making some people a lesser man and others a better man”.

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[2]

[Total: 13]

- 3** In 2006, scientists Takahashi and Yamanaka stimulated mouse tail-tip cells, which are fully differentiated cells, to change back into stem cells in tissue culture. Such stem cells obtained from the de-differentiation of somatic cells are known as induced pluripotent stem cells (iPS cells). In the last decade, iPS cells have been generated from humans cells isolated from different tissues by introduction of various growth factors that affects the expression of many genes, such as the gene coding for human telomerase reverse transcriptase (hTERT).

(a) Explain how the expression of hTERT may help the cells regain stem cell properties.

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[3]

- (b)** X-linked Severe Combined Immunodeficiency (SCID) is a rare congenital disorder characterised by improper development of immune cells which has been treated by gene therapy.

Explain why X-linked SCID can be treated by gene therapy.

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[2]

The ability to generate iPS cells that have characteristics that are similar to embryonic stem cells has provided a promising alternative to the use of haematopoietic stem cells for gene therapy to treat X-linked SCID. Fig 3.1 shows a possible process of using iPS cells for gene therapy.

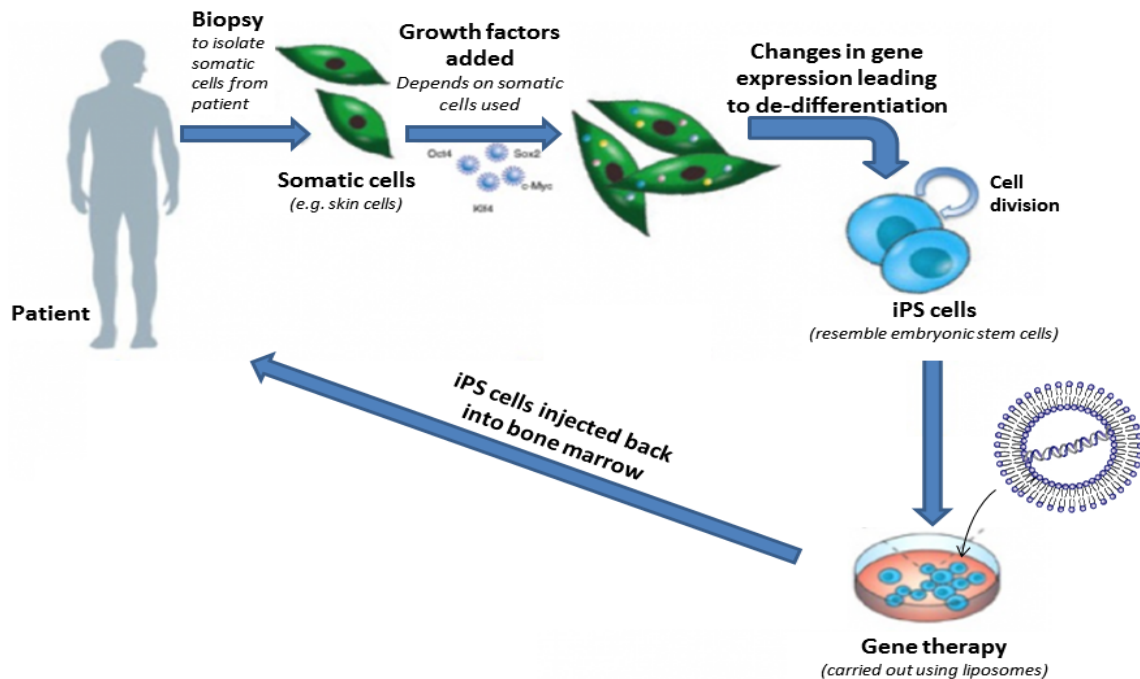


Fig. 3.1

- (c) Describe and explain **two** factors that prevent this method of gene therapy for X-linked SCID from becoming an effective treatment.

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[4]

- (d) Although the use of retroviral vectors for gene therapy has been highly successful in treating SCID, many patients who received treatment also developed immune responses against the virus and leukaemia.

Suggest another challenge of using viral vectors in the process shown in Fig. 3.1.

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[1]

- (e) (i) Discuss the arguments for and against the use of human embryonic stem cells for therapy.

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..... [2]

- (ii) Explain whether you agree or disagree that use of iPS cells resolve the issues discussed above.

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..... [1]

[Total: 13]

4 Planning question

Diffusion of molecules can be observed using the Visking tubing.

The Visking (dialysis) tubing is a selectively permeable membrane that allows small molecules to diffuse through. The tubing can be tightly knotted at one end and a solution can be contained within the tubing. The filled tubing can then be placed in distilled water for the diffusion of the solute molecules out of the tubing into the distilled water. Fig. 4.1 shows the Visking tubing set-up as described.

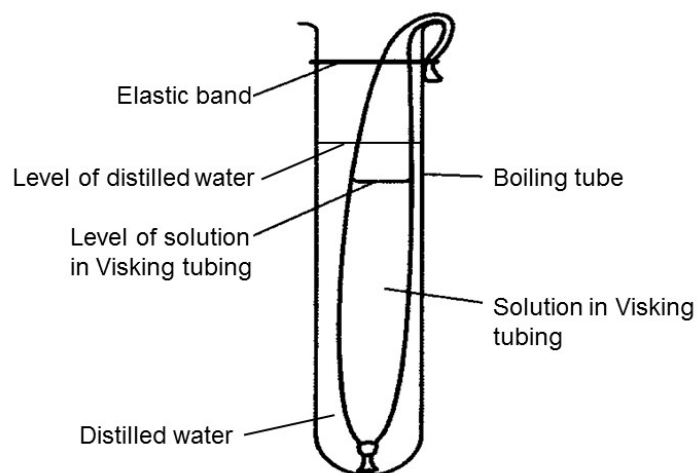


Fig. 4.1

The presence of solutes in distilled water surrounding the tubing can then be tested using quantitative / qualitative analysis.

Presence of glucose in solution can be tested using Benedict's test. The concentration of glucose in a solution can be estimated using a colour standard made with solutions with known glucose concentrations.

Using this information and your own knowledge, design an experiment to investigate the effect of temperature on the diffusion of glucose.

You must use:

- 100 cm³ of 10g / 100 cm³ glucose solution,
- 10 cm³ of 5g / 100 cm³ glucose solution for colour standard,
- Distilled water,
- Thermostatically-controlled water bath and thermometer,
- Benedict's solution,
- Boiling tubes,
- Stopwatch.

You may select from the following apparatus and chemicals:

- Normal laboratory glassware, e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods etc.,
- Syringes.

Your plan should:

- Have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it,
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results and the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimise any risks associated with the proposed experiment.

[Total: 12]

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Free-response question

Write your answers to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5 **(a)** Outline nucleic acid hybridisation and explain how it can be used in the detection of DNA sequences. [9]
- (b)** Explain the advantages and disadvantages of tissue culture over more traditional methods of cloning plants, such as taking cuttings or grafting. [5]
- (c)** Describe the goals of the Human Genome Project, and with a named example, explain how the findings of the HGP may benefit the genetic modification of crops. [6]

[Total:20]